

KUNAL M SHENDE

Software Engineer | IoT Specialist | Ethical Hacker

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SUMMARY

Innovative Computer Science undergraduate with over 2 years of proven expertise across embedded hardware, system hardening, and full-stack software development. Pioneered ZeroTrace, an ESP32-S3 audit companion delivering real-time RF spectrum analysis and USB automation, earning 2nd Prize among 50+ teams at the TECHTANTRA national competition and securing an official Indian Design Patent filing (Application No: 499678-001). Adept at penetration testing, vulnerability mitigation, and deploying modern web applications using React.js, Next.js, Node.js, and Python.

SKILLS

Languages: C, Embedded C, C++, Python, JavaScript, TypeScript, Bash

Hardware & Embedded: ESP32, ESP32-S3, Arduino, Raspberry Pi, FreeRTOS, SPI, I2C, UART

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, Bootstrap

Backend: Python Flask, Node.js, Express.js, WebSockets, Firebase, REST APIs

Databases: MySQL, PostgreSQL, Supabase, Firestore, SQLite

Cyber Defense Tools: VAPT, Burp Suite, Nmap, Wireshark, Metasploit, OWASP Top 10

DevOps & Utilities: Git, GitHub, Docker, Vercel, Google Cloud Platform, Postman, VS Code

EXPERIENCE

Independent Security Researcher | Self-Employed | Jan 2024 - Present

- Architected 3 micro-controller prototypes, decreasing firmware attack exposure by 40% via hardware bus isolation techniques.
- Identified 15+ high-risk system vulnerabilities across 10+ lab environments following NIST assessment protocols.
- Lowered unauthorized data anomalies by 25% by implementing continuous packet inspection on 5 target devices.
- Authored 3 threat mitigation whitepapers assisting embedded firmware engineers with proactive defense strategies.

Vulnerability Assessment Trainee | EduSkills Foundation | Jun 2025 - Aug 2025

- Discovered 12 critical application weaknesses including SQL injection and XSS across 8 client test platforms.
- Accelerated software remediation timelines by 30% by delivering structured proof-of-concept validation reports.

SecOps & Defense Trainee | Palo Alto Networks | Jan 2025 - Mar 2025

- Examined 500+ packet captures and firewall policies, reinforcing perimeter safety across 3 virtual server environments.
- Completed 4 professional credentials validating proficiency in threat mitigation and incident response.

Java Full Stack Developer | AICTE Virtual Internship | Oct 2024 - Dec 2024

- Constructed 2 full-stack portals with Spring Boot backend logic connecting relational schemas via 15+ REST endpoints.
- Optimized query retrieval latency by 20% through relational index refactoring, enhancing experience for 500+ active users.

Cloud Infrastructure Trainee | AWS Academy | Apr 2024 - Jun 2024

- Provisioned 10+ cloud infrastructure resources including EC2, S3, VPC, and IAM role governance.
- Shortened manual environment setup time by 60% utilizing CloudFormation infrastructure-as-code scripts.

Mobile Application Developer | Google for Developers | Jan 2024 - Mar 2024

- Delivered 2 native Android applications integrating Material Design components and offline SQLite storage.
- Maintained 99.5% operational uptime while processing 1,000+ daily client synchronization transactions.

PROJECTS

ZeroTrace - Hardware Audit Companion | ESP32-S3, FreeRTOS, C++, USB-HID | 2025 - 2026

- Engineered a portable auditing device featuring 2.4 GHz spectrum scanning, wireless probing, and keystroke injection.
- Achieved smooth 60 FPS graphical rendering on a color display by distributing tasks across dual CPU cores, cutting latency by 35%.
- Won 2nd Prize at the TECHTANTRA national tournament and filed Indian Design Patent No. 499678-001.

Sentinel - Host Intrusion Detection System | *Python, Windows API, WMI* | 2025 - 2026

- Built a real-time host protection utility that flags unauthorized keystroke injection with sub-10ms response speed.
- Neutralized 100% of simulated BadUSB attack vectors through automated process termination and digital snapshotting.

MicroFlasher - Web Serial Firmware Platform | *TypeScript, Next.js, Web Serial API* | 2025 - 2026

- Launched an in-browser firmware programmer, eliminating local toolchain installs and cutting onboarding time by 70%.
- Applied encrypted binary protection supporting microcontroller uploads up to 4 MB with live progress monitoring.

Autonomous Line-Follower Robot | *ATmega328P, L298N, IR Sensors, C++* | 2024 - 2025

- Assembled a differential-drive robot with PWM timer tuning at 7.8 kHz, eliminating motor stall and reducing whine by 90%.
- Tuned a closed-loop proportional controller achieving 95% path-tracking accuracy on sharp 90-degree corners.

Chai & Kafe - Artisan Bistro Portal | *HTML5, CSS3, JavaScript, Bootstrap* | 2024 - 2025

- Delivered a responsive restaurant web platform with interactive ordering and table reservations, achieving a 95+ Lighthouse score.

AWARDS & ACHIEVEMENTS

- Secured 2nd Prize at TECHTANTRA 2K26 National Technical Project Competition among 50+ participating collegiate teams.
- Filed Indian Design Patent (Application No: 499678-001, CBR NO: 209757) in April 2026 for ZeroTrace hardware enclosure.

EDUCATION

B.Tech in Computer Science & Engineering | *S. B. Jain Institute of Technology, Nagpur* | 2023 - 2027

- Relevant Coursework: Operating Systems, Computer Networks, Data Structures, Embedded Systems, and IoT Security.

Higher Secondary Certificate (HSC) - Science | *Municipal Junior College, Kalmeshwar* | 2021 - 2023

- Curriculum in Mathematics, Physics, and Chemistry under Maharashtra State Board.

Secondary School Certificate (SSC) - Distinction | *Municipal Junior College, Kalmeshwar* | 2019 - 2021

- Graduated with Distinction honor under Maharashtra State Board.

CERTIFICATIONS

- Ethical Hacking and Penetration Testing - NASSCOM FutureSkills Prime.
- Cyber Security Workshop Credential - Crow Security & IIT Bombay.
- Security Operations (SecOps) Fundamentals - Palo Alto Networks.
- Network Security & Cloud Defense Fundamentals - Palo Alto Networks.
- Cloud Computing Architecture Graduate - AWS Academy.
- Python Problem Solving Certification - HackerRank.
- German Language Proficiency Certificate (Level 1) - FLOA.

LANGUAGES

English (Professional Working), Hindi (Native), Marathi (Native), German (Elementary A1)